Paper Summary

Junseok Lee

August 3, 2026

1 Flexible Parametric Approach to Classical Measurement Error Variance Estimation without Auxiliary Data

1.1 The problem

Measurement error in the continuous covariates of a model yields bias in the estimators. In most cases, many correction methods require additional information about measurement error distributions. However, neither validation nor auxiliary data (e.g., replicated measurements) are available.
Therefore, this paper considers how to estimate the variance of classical additive Gaussian measurement error without auxiliary data. The main goal is to estimate the unknown error standard deviation v from the observed contaminated covariate W alone, under the weak assumption that the true covariate X has compact support.

1.2 The setup

The paper considers the classical additive measurement error model on

$$W = X + U,$$

where W is the observed covariate, X is the true but unobservable covariate, and U is a mean-zero measurement error satisfying

$$U \sim N(0, v^2), \qquad X \perp\!\!\!\perp U.$$

The error standard deviation v is unknown. The distribution of X is also unknown, **but X is assumed to be continuous and compactly supported.**

To express the compact support flexibly, the paper assumes

$$X = aS + b,$$

where $a > 0$, $b \in \mathbb{R}$, and S is a continuous random variable taking values in $[0, 1]$. Thus the density of W can be written as

$$f_W(w) = \frac{1}{v} \int_b^{a+b} f_X(x) \phi \left(\frac{w - x}{v} \right) dx = \frac{1}{av} \int_b^{a+b} f_S \left(\frac{x - b}{a} \right) \phi \left(\frac{w - x}{v} \right) dx.$$

where f_S is the density of S and ϕ is the standard normal density. **The goal is to estimate**

$$(v, a, b, f_S)$$

based only on i.i.d. observations

$$W_1, \ldots, W_n \overset{\text{iid}}{\sim} W.$$

1.3 The main idea

The key idea is **to approximate the unknown density f_S by Bernstein polynomials.** Since any continuous function on $[0,1]$ can be **uniformly approximated** by Bernstein polynomials, i.e.,

$$\lim_{m\to\infty} \sup_{0\le s\le 1} \left| \sum_{k=0}^{m} f\left(\frac{k}{m}\right) b_{k,m}(s) - f(s) \right| = 0.$$

this gives a flexible parametric approximation of the unknown density of S.

$$f_S \implies \widetilde{f}_{S,m} \implies \widetilde{f}_{X,m} \implies \widetilde{f}_{W,m} \implies \text{likelihood for } W_1, \ldots, W_n.$$

For a fixed degree m, f_S is approximated by a mixture of Beta densities:

$$\widetilde{f}_{S,m}(s;\theta_m) = \sum_{k=0}^{m} f_S\left(\frac{k}{m}\right) b_{k,m}(s) = \sum_{k=0}^{m} \theta_{k,m} \frac{(m+1)!}{k!(m-k)!} s^k (1-s)^{m-k}, \qquad s \in [0,1].$$

Equivalently,

$$\widetilde{f}_{S,m}(s;\theta_m) = \sum_{k=0}^{m} \theta_{k,m}\, \mathrm{Beta}_{k+1,m-k+1}(s),$$

where $\mathrm{Beta}_{\alpha,\beta}$ denotes the density of a Beta distribution with parameters α and β.

Then the density of $X = aS + b$ is approximated by

$$\widetilde{f}_{X,m}(x;a,b,\theta_m) = \frac{1}{a} \sum_{k=0}^{m} \theta_{k,m}\, \mathrm{Beta}_{k+1,m-k+1}\left(\frac{x-b}{a}\right), \qquad b \le x \le a+b.$$

Since $W = X + U$, the density of W is approximated by the convolution

$$\widetilde{f}_{W,m}(w;v,a,b,\theta_m) = \frac{1}{av} \sum_{k=0}^{m} \theta_{k,m} \int_b^{a+b} \mathrm{Beta}_{k+1,m-k+1}\left(\frac{x-b}{a}\right) \phi\left(\frac{w-x}{v}\right) dx.$$

Therefore, the log-likelihood is

$$L_n(v,a,b,\theta_m) = \sum_{i=1}^{n} \log \left[\frac{1}{av} \sum_{k=0}^{m} \theta_{k,m} \int_b^{a+b} \mathrm{Beta}_{k+1,m-k+1}\left(\frac{x-b}{a}\right) \phi\left(\frac{W_i-x}{v}\right) dx \right].$$

For each fixed m, the parameters are estimated by maximizing this likelihood:

$$(\widehat{v}_m, \widehat{a}_m, \widehat{b}_m, \widehat{\theta}_m) = \arg\max_{v,a,b,\theta_m} L_n(v,a,b,\theta_m).$$

The degree m controls the flexibility of the density approximation. A larger m gives a more flexible density approximation, but it also introduces more parameters. The reason of using BIC rather than AIC is that it chooses more simpler models, so the estimation of the measurement error v is more stable.

Therefore, the paper chooses m by BIC:

$$\mathrm{BIC}(m) = (m+3)\log n - 2L_n(\widehat{v}_m, \widehat{a}_m, \widehat{b}_m, \widehat{\theta}_m).$$

Thus, the final estimator of the measurement error standard deviation is

$$\widehat{v} = \widehat{v}_{\widehat{m}}, \qquad \widehat{m} = \arg\min_m \mathrm{BIC}(m).$$

1.4 The main results

The first main theoretical result is identifiability. Under the Gaussian classical measurement error model and the compact support assumption on X, there exist unique a, b, v^2, and f_S such that the density representation of W holds. Hence, the measurement error variance is identifiable from the marginal distribution of W.

For fixed m, the proposed estimator is treated as a maximum likelihood estimator under a possibly misspecified parametric model. Then, under regularity conditions,

$$(\widehat{v}_m, \widehat{a}_m, \widehat{b}_m, \widehat{\theta}_m) \xrightarrow{p} (v_m^*, a_m^*, b_m^*, \theta_m^*).$$

The paper also establishes asymptotic normality:

$$\sqrt{n}\left\{(\widehat{v}_m, \widehat{a}_m, \widehat{b}_m, \widehat{\theta}_m) - (v_m^*, a_m^*, b_m^*, \theta_m^*)\right\} \xrightarrow{d} N(0, C),$$

Simulation studies show that the proposed estimator works well for many compactly supported distributions of X. However, the method can have practical identifiability problems when the density of X has thin tails, because the estimation of v is closely related to the estimation of the support of X.

1.5 My takeaways

- This method just investigate in the compact support, what if the support is not compact? How to deal it?
  A: To use the uniform approximation of the Bernstein polynomial. But I don't know how to apply more general support.
- Why this method rely on BIC rather than AIC? What is the purpose? for stable estimation of the v?
- Why the author assumes the U's distribution is Gaussian and tries to estimate the variance rather than other statistical quantity?
- Why does this paper further estimate the variance of U, even though it estimates the density f_X?
  A: For complete estimation, it should identify the distribution of U and check whether $W = X + U$ is properly constructed.

2 Nonparametric Estimation on the Circle Based on Fejér Polynomials

2.1 The problem

Nonparametric density estimation and distribution function estimation are well developed for Euclidean data, but circular data require methods that deal with periodicity. Some previous approaches adapt Bernstein polynomial estimators from the unit interval to the circle. However, such approaches do not naturally exploit the periodic structure of circular densities.

Therefore, this paper develops nonparametric density and CDF estimators on the circle using Fejér polynomials. Fejér polynomials are natural analogues of Bernstein polynomials for periodic functions. The paper also extends the proposed estimators to measurement error settings, including both Berkson and classical error models.

2.2 The setup

Let

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} X$$

be circular observations with density f and CDF F on $[-\pi, \pi)$. The goal is to estimate f and F nonparametrically while respecting the periodic nature of the sample space.

The Fejér kernel of order m is defined by

$$K_m(s) = \frac{1}{2\pi} \sum_{k=-m}^{m} \left(1 - \frac{|k|}{m+1}\right) e^{iks} = \frac{1}{2\pi} + \frac{1}{\pi} \sum_{k=1}^{m} \left(1 - \frac{k}{m+1}\right) \cos(ks).$$

It is continuous, symmetric, periodic, nonnegative, and integrates to one:

$$\int_{-\pi}^{\pi} K_m(s)\, ds = 1.$$

Thus, K_m can be used as a circular smoothing kernel.

The paper also considers measurement error models on the circle. In the Berkson error model,

$$X = (X^\star + \varepsilon) \mod 2\pi,$$

where $X^\star$ is observed and X is the target variable. In the classical error model,

$$X^\star = (X + \varepsilon) \mod 2\pi,$$

where $X^\star$ is observed and X is unobservable. The error distribution is assumed to be known.

2.3 The main idea

The key idea is to use Fejér's theorem for periodic functions. If f is a continuous periodic density, then f can be uniformly approximated by its Fejér polynomial:

$$\sigma_m(f; \theta) = \frac{1}{2\pi} \sum_{k=-m}^{m} \left(1 - \frac{|k|}{m+1}\right) \phi_k e^{ik\theta},$$

where

$$\phi_k = \int_{-\pi}^{\pi} f(t) e^{-ikt}\, dt$$

is the k-th Fourier coefficient of f.

Since ϕ_k is unknown, the paper replaces it with the empirical Fourier coefficient

$$\widehat{\phi}_{n,k} = \frac{1}{n} \sum_{j=1}^{n} e^{-ikX_j}.$$

This gives the Fejér density estimator

$$\widehat{f}_{m,n}(\theta) = \frac{1}{2\pi} \sum_{k=-m}^{m} \left(1 - \frac{|k|}{m+1}\right) \widehat{\phi}_{n,k} e^{ik\theta}.$$

Equivalently, this estimator can be written as a kernel density estimator:

$$\widehat{f}_{m,n}(\theta) = \frac{1}{n} \sum_{j=1}^{n} K_m(\theta - X_j).$$

Thus, the estimator is obtained by replacing the unknown Fourier coefficients in the Fejér approximation by empirical Fourier coefficients.

For CDF estimation, the paper integrates the Fejér density estimator. Let

$$W_m(\theta) = \int_{-\pi}^{\theta} K_m(y)\,dy = \frac{\theta+\pi}{2\pi} + \frac{1}{\pi}\sum_{k=1}^{m}\left(1-\frac{k}{m+1}\right)\frac{\sin(k\theta)}{k}.$$

Then the Fejér CDF estimator is

$$\widehat{F}_{m,n}(\theta) = \int_{-\pi}^{\theta} \widehat{f}_{m,n}(y)\,dy = \frac{1}{n}\sum_{i=1}^{n}\{W_m(\theta-X_i) - W_m(-\pi-X_i)\}.$$

A special issue for circular CDF estimation is that the CDF depends on the choice of the origin. For an origin θ_0, define

$$F^{\theta_0}(\theta) = \int_{\theta_0}^{\theta} f(y)\,dy.$$

The paper proposes choosing θ_0 by minimizing an empirical counterpart of the leading variance term. If the data are transformed to $[\theta_0, \theta_0+2\pi)$ and ordered as

$$x_{(1)}^{\theta_0} \le \cdots \le x_{(n)}^{\theta_0},$$

then the empirical criterion is

$$C_n(\theta_0) = \sum_{i=1}^{n}\frac{i}{n}\left(1-\frac{i}{n}\right)\left(x_{(i+1)}^{\theta_0} - x_{(i)}^{\theta_0}\right).$$

The estimated origin is chosen by minimizing $C_n(\theta_0)$ over $\theta_0 \in [-\pi, \pi)$.

For measurement error, the paper uses Fourier deconvolution. If the error density has Fourier coefficients λ_ℓ, then the Berkson estimator multiplies by the error coefficients:

$$\widehat{f}_{\text{Berk}}(x;m) = \frac{1}{2\pi n}\sum_{j=1}^{n}\left[1 + 2\sum_{\ell=1}^{m}\left(1-\frac{\ell}{m+1}\right)\lambda_\ell\cos\{\ell(x-X_j^\star)\}\right].$$

In contrast, the classical error estimator divides by the error coefficients:

$$\widehat{f}_{\text{class}}(x;m) = \frac{1}{2\pi n}\sum_{j=1}^{n}\left[1 + 2\sum_{\ell=1}^{m}\frac{1-\ell/(m+1)}{\lambda_\ell}\cos\{\ell(x-X_j^\star)\}\right].$$

2.4 The main results

The first main result is the uniform strong consistency of the Fejér density estimator. This follows from Fejér's uniform approximation theorem and uniform convergence of the empirical Fourier coefficients.

The paper also derives the MISE expansion of the density estimator. If f is not the circular uniform density, the asymptotically optimal order of m is $m_{\text{opt}} = (6\pi\theta_1(f))^{1/3}n^{1/3}$.

For the measurement error extensions, the paper shows how the Fejér estimator can be modified in both Berkson and classical error models. In the Berkson case, the optimal bandwidth can be chosen as in the error-free case. In the classical case, deconvolution increases the variance.

2.5 My takeaways

3 Deconvolution density estimation on Lie groups without auxiliary data

3.1 The problem

This paper studies deconvolution density estimation when the data lie on a Lie group and are contaminated by measurement error. In the usual Euclidean measurement error setting, one observes

$$Z = X + U,$$

where X is the unobserved target variable and U is an unobserved measurement error. If the distribution of U is known, then the density of X can be estimated by deconvolution. However, in many situations, the distribution of U is not known.

This paper considers the difficult case where there is no auxiliary data. To make the problem identifiable, the paper takes a semiparametric approach: the measurement error distribution is assumed to belong to a parametric family, while the target distribution of X is treated nonparametrically.

Although the paper is written for a general Lie group $\mathbb{G}$, the most relevant special case for us is

$$\mathbb{G} = \mathbb{T}^1 \simeq \mathbb{S}^1,$$

where the group operation corresponds to addition modulo 2π. Therefore, in the circular case, the model can be understood as

$$\theta_Z = \theta_U + \theta_X \pmod{2\pi}.$$

3.2 The setup

Let $X \in \mathbb{G}$ be the target random element and let $U \in \mathbb{G}$ be an unobservable measurement error independent of X. The observed random element is

$$Z = U \circ X.$$

For $\mathbb{G} = \mathbb{T}^1$, write

$$X = e^{i\theta_X}, \qquad U = e^{i\theta_U}, \qquad Z = e^{i\theta_Z},$$

where $\theta_X, \theta_U, \theta_Z \in [0, 2\pi)$. Then the group operation becomes

$$\theta_Z = \theta_U + \theta_X \pmod{2\pi}.$$

The goal is to estimate the distribution of X and the distribution of U using only

$$Z_1, \ldots, Z_n.$$

In particular, the paper assumes that the distribution of U belongs to a parametric family. In the compact Lie group case, this family is written in terms of Fourier transforms as

$$\phi_U(\sigma^M) = \exp(-\gamma_0 k_\sigma^\beta) I_{d_\sigma},$$

where $\gamma_0 > 0$ is the unknown measurement error parameter, $\beta > 0$ is known, and k_σ is the eigenvalue index associated with the representation σ.

For $\mathbb{G} = \mathbb{T}^1$, the irreducible representations are indexed by $\ell \in \mathbb{Z}$, and

$$\sigma_\ell^M(g) = e^{-i\ell\theta_g}, \qquad k_{\sigma_\ell} = \ell^2.$$

Thus the measurement error family becomes

$$\phi_U(\ell) = \exp(-\gamma_0 |\ell|^{2\beta}).$$

For example, the wrapped Gaussian distribution satisfies

$$\phi_U(\ell) = \exp\left(-\frac{\lambda_U^2 \ell^2}{2}\right),$$

so it corresponds to

$$\beta = 1, \qquad \gamma_0 = \frac{\lambda_U^2}{2}.$$

3.3 The main idea

The key identity is the convolution identity. Since

$$Z = U \circ X$$

and U is independent of X, we have

$$P_Z = P_U * P_X.$$

On the Fourier side, this gives

$$\phi_Z(\sigma^M) = \phi_X(\sigma^M)\phi_U(\sigma^M).$$

In the special case $\mathbb{G} = \mathbb{T}^1$, this becomes

$$\phi_Z(\ell) = \phi_X(\ell)\phi_U(\ell).$$

Under the parametric measurement error family,

$$\phi_U(\ell) = \exp(-\gamma_0 |\ell|^{2\beta}),$$

so

$$|\phi_Z(\ell)| = |\phi_X(\ell)| \exp(-\gamma_0 |\ell|^{2\beta}).$$

The main identifiability idea is to look at high frequencies. If the Fourier coefficients of X do not decay as fast as the measurement error coefficients, then the leading exponential decay of $|\phi_Z(\ell)|$ is dominated by U. In other words,

$$\frac{\log |\phi_Z(\ell)|}{|\ell|^{2\beta}} = \frac{\log |\phi_X(\ell)|}{|\ell|^{2\beta}} - \gamma_0.$$

Under the assumptions on P_X, the first term satisfies

$$\frac{\log |\phi_X(\ell)|}{|\ell|^{2\beta}} \longrightarrow 0,$$

and therefore

$$\frac{\log |\phi_Z(\ell)|}{|\ell|^{2\beta}} \longrightarrow -\gamma_0.$$

This means that the measurement error parameter γ_0 can be identified from the high-frequency decay rate of the Fourier coefficients of Z.

This leads to the estimator

$$\widehat{\gamma}_0 = -\frac{\log \left| n^{-1} \sum_{i=1}^{n} \exp\{is(n)\theta_{Z_i}\} \right|}{s(n)^{2\beta}},$$

where $s(n) \to \infty$ is a frequency index. Once γ_0 is estimated, the Fourier coefficients of X can be recovered by

$$\widehat{\phi}_X(\ell) = \widehat{\phi}_Z(\ell) \exp(\widehat{\gamma}_0 |\ell|^{2\beta}).$$

The density estimator is then obtained by truncated Fourier inversion. For $\mathbb{G} = \mathbb{T}^1$, the estimator can be written as

$$\widehat{f}_X(x) = \sum_{\ell : \ell^2 < T_n} \exp(\widetilde{\gamma}_0 \ell^{2\beta}) \frac{1}{n} \sum_{i=1}^{n} \cos\{\ell(\theta_{Z_i} - \theta_x)\},$$

where $T_n \to \infty$ is a tuning parameter and $\widetilde{\gamma}_0$ is an estimator of γ_0. The truncation is necessary because

$$\exp(\widetilde{\gamma}_0 \ell^{2\beta})$$

diverges as $|\ell| \to \infty$, so using all Fourier frequencies would make the estimator unstable.

Thus, the overall procedure is

$$Z_1, \ldots, Z_n \Longrightarrow \widehat{\phi}_Z(\ell) \Longrightarrow \widehat{\gamma}_0 \Longrightarrow \widehat{\phi}_U(\ell) \Longrightarrow \widehat{\phi}_X(\ell) \Longrightarrow \widehat{f}_X.$$

3.4 The main results

The first main result is identifiability. Under the assumed parametric family for P_U and suitable conditions on P_X, the map

$$(P_U, P_X) \mapsto P_U * P_X$$

is injective. This means that P_U and P_X are uniquely determined by the distribution of Z. In the $\mathbb{T}^1$ case, this means that the observed circular distribution of Z contains enough information to identify both the measurement error parameter γ_0 and the target distribution P_X, provided that the Fourier coefficients of X do not decay too fast compared with those of U.

The paper next studies estimation of γ_0. Under condition (X1), the estimator satisfies

$$\widehat{\gamma}_0 - \gamma_0 = O_P\left(\frac{\log \log n}{\log n}\right).$$

Under the broader condition (X2), the convergence rate is slightly slower. This is natural because (X2) allows a wider class of target distributions P_X.

After estimating γ_0, the paper constructs a deconvolution density estimator for f_X. For compact Lie groups, including $\mathbb{T}^1$, the estimator is based on a truncated Fourier series. The paper derives both L^2 convergence rates and uniform convergence rates. In the compact group case, the L^2 error rate has three main components:

$$\text{bias from truncation} + \text{error from estimating } \gamma_0 + \text{variance from estimating } \phi_Z.$$

The finite truncation level T_n balances these terms and prevents instability from the exponentially increasing deconvolution factor.

The paper also considers the fully parametric case, where both P_U and P_X belong to parametric families. In that case, maximum likelihood estimation can be used, and the paper establishes consistency and asymptotic normality of the maximum likelihood estimator.

Finally, the simulation study focuses on

$$\mathbb{G} = \mathbb{T}^1.$$

The paper considers wrapped Gaussian measurement error and target distributions such as wrapped Laplace and wrapped Cauchy. The results show that the proposed estimators improve as the sample size increases. The simulation also shows that the choice of the high-frequency index $s(n)$ is important, and selecting it by least squares cross-validation performs better than using a fixed theoretically motivated choice.

3.5 My takeaways